Inorganic CUME
January 23, 2015

1. In search for new battery materials, often people need to look at the reduction half reaction emf of an element or its compounds. Which element's reduction half reaction gives the most positive *emf*, and which gives the most negative *emf*? (*10 points*)

2. When designing high-temperature fuel-cells, one might need to know which elements (or their compounds) would offer the highest metaling point. Identify the element(s) having the highest melting point in each row of the transition metals. (*15 points*)

3. In study of cation-based rechargeable batteries, we may need to know the cation's radii. Discuss trends in radii of naked and hydrated cations for the alkali metals and alkaline earth metals. (*20 points*)

4. Chemical corrosion has been a longstanding issue in materials degradation. Rank each group of the chemicals in an order of increasing acidity. (*10 points*)
   i) SiH_4, CH_4, GeH_4
   ii) Al_2O_3, MgO, Na_2O, SiO_2

5. In developing new science and technology for renewable energy, semiconductors are commonly seen. Please show a reaction for making a semiconducting chalcogenide, draw the crystal structure's unit cell for the semiconductor. (*20 points*)

6. Chemistry and structures of pure carbon-based molecules and materials: (*25 points*)
   (i). Indicate which carbon structure(s) would be conductive, semiconducting, insulating.
   (ii) Explain why the carbonaceous species show the characteristic colors.